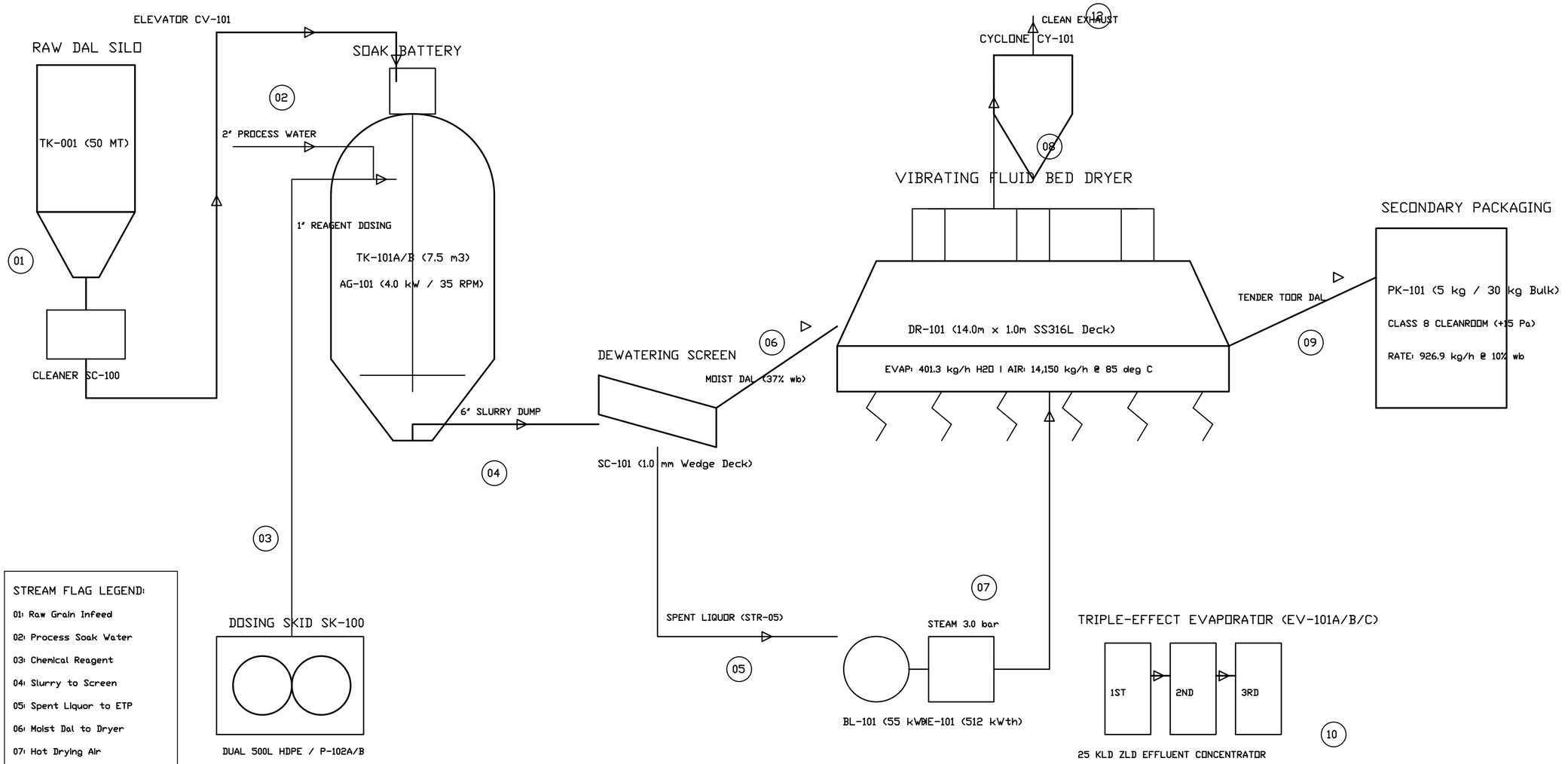
KEY PROCESS DESIGN BASIS & THERMODYNAMIC OPERATING PARAMETERS:

Throughput: 15,000 kg/day as-fed Dal | Operating Schedule: 16.0 h/day (Mean 937.5 kg/h, Design 1,102.9 kg/h @ 85% OEE)

Infeed Moisture: 10.50% w.b. (839.06 kg/h dry solids) | Product Moisture: 10.00% w.b. (926.90 kg/h saleable dal) | Solids Yield: 99.40%

Soak Liquor Ratio: 1.50 : 1.00 (Water:Dal) | Dosing Reagent: Food-Grade NaHCO3 / Citrate | Drying Air: 85 deg C | Deck Area: 14.0 m2

All Product-Contact Metallurgy: AISI SS316L 2B Sanitary Polish (Ra <= 0.8 um) | CIP / Washdown Compatible | ZLD Effluent Evaporation



STREAM FLAG LEGEND:

- 01: Raw Grain Infeed
- 02: Process Soak Water
- 03: Chemical Reagent
- 04: Slurry to Screen
- 05: Spent Liquor to ETP
- 06: Moist Dal to Dryer
- 07: Hot Drying Air
- 09: Saleable Toor Dal

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.		
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project		
PROCESS FLOW DIAGRAM - OVERALL PLANT		
DISCIPLINE: PROCESS | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)		
DWG NO: DWG-FIN-15T-001	REV: V8.0	SHEET: 1 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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STR	STREAM DESCRIPTION	SOURCE	DESTINATION	TOTAL (kg/h)	SOLIDS (kg/h)	WATER (kg/h)	MOIST (%wb)	T (degC)	P (bar g)	PHASE	VOL (m3/h)
01	Raw Split Pigeon Pea Infeed	Silo TK-001	Elevator CV-101	937.50	839.06	98.44	10.50%	28.0	ATM	Solid Granular	1.17
02	Treated Soft Process Water	Water Plant WTP-01	Soak Battery TK-101	1,406.25	0.00	1,406.25	100.00%	30.0	2.00	Liquid	1.41
03	Reagent Dosing Solution (10%)	Dosing Skid SK-100	Soak Battery TK-101	14.06	1.41	12.65	90.00%	28.0	3.00	Aqueous Soln	0.01
04	Hydrated Dal Slurry Discharge	Soak Battery TK-101	Screen SC-101	2,357.81	840.47	1,517.34	64.35%	30.0	0.20	Solid-Liquid	2.14
05	Separated Spent Soak Liquor	Screen SC-101	Equalization TK-104	1,018.75	6.44	1,012.31	99.37%	30.0	ATM	Turbid Liquid	1.02
06	Moist Dewatered Dal to Dryer	Screen SC-101	Fluid Bed DR-101	1,339.06	834.03	505.03	37.72%	30.0	ATM	Wet Particulate	1.58
07	Hot Conditioned Drying Air	Air Heater HE-101	Plenum DR-101	14,150.00	0.00	155.65	1.10% db	85.0	0.03	Gas	11,800.0
08	Moist Vapour Exhaust Air	Dryer Hood DR-101	Cyclone CY-101	14,551.31	10.60	556.96	3.83% db	62.0	-0.01	Dusty Gas	12,950.0
09	Finished Tender Toor Dal	Dryer DR-101	Packaging PK-101	926.90	834.21	92.69	10.00%	35.0	ATM	Solid Granular	1.16
10	Evaporator Clean Condensate	Condenser HE-103	Water Tank TK-105	980.00	0.00	980.00	100.00%	45.0	ATM	Distillate Liq	0.99
11	ZLD Concentrate Effluent	3rd Effect EV-101C	Solar Pan / Bagging	38.75	15.50	23.25	60.00%	55.0	0.50	Heavy Slurry	0.03
12	Cyclone Separated Fines	Cyclone CY-101	Packaging Recycle	10.60	9.54	1.06	10.00%	58.0	ATM	Fine Powder	0.02

MASS BALANCE RECONCILIATION & SOLIDS CONSERVATION AUDIT:

1. DRY SOLIDS CONSERVATION CHECK: Exact 0.0000% mathematical closure across entire flowsheet.
   - Raw Grain Solids Infeed (Stream 01): 839.06 kg/h (100.000% basis).
   - Chemical Reagent Active Minerals (Stream 03): 1.41 kg/h.
   - Product Saleable Solids Output (Stream 09): 834.21 kg/h (99.42% solids yield).
   - Spent Liquor Soluble Losses (Stream 05): 6.26 kg/h (0.75% leaching).
   - Total Output Solids = 834.21 + 6.26 = 840.47 kg/h == Total Input Solids (839.06 + 1.41 = 840.47 kg/h).
2. WATER EVAPORATION DUTY IN FLUID BED DRYER DR-101:
   - Moisture Inflow (Stream 06): 505.03 kg/h | Moisture Outflow (Stream 09): 92.69 kg/h.
   - Net Water Evaporation Rate: 412.34 kg/h H2O @ 85 deg C drying air.
3. ZERO LIQUID DISCHARGE (ZLD) EFFLUENT RECOVERY:
   - Spent Liquor Inflow to Evaporator: 1,018.75 kg/h (99.37% water).
   - Distillate Clean Water Recycled (Stream 10): 980.00 kg/h (96.20% volumetric recovery).
   - Heavy Mineral Concentrate Bleed (Stream 11): 38.75 kg/h (40.0% solids slurry to solar pan).

THERMAL ENERGY, STEAM CONSUMPTION & ELECTRICAL POWER SCHEDULE:

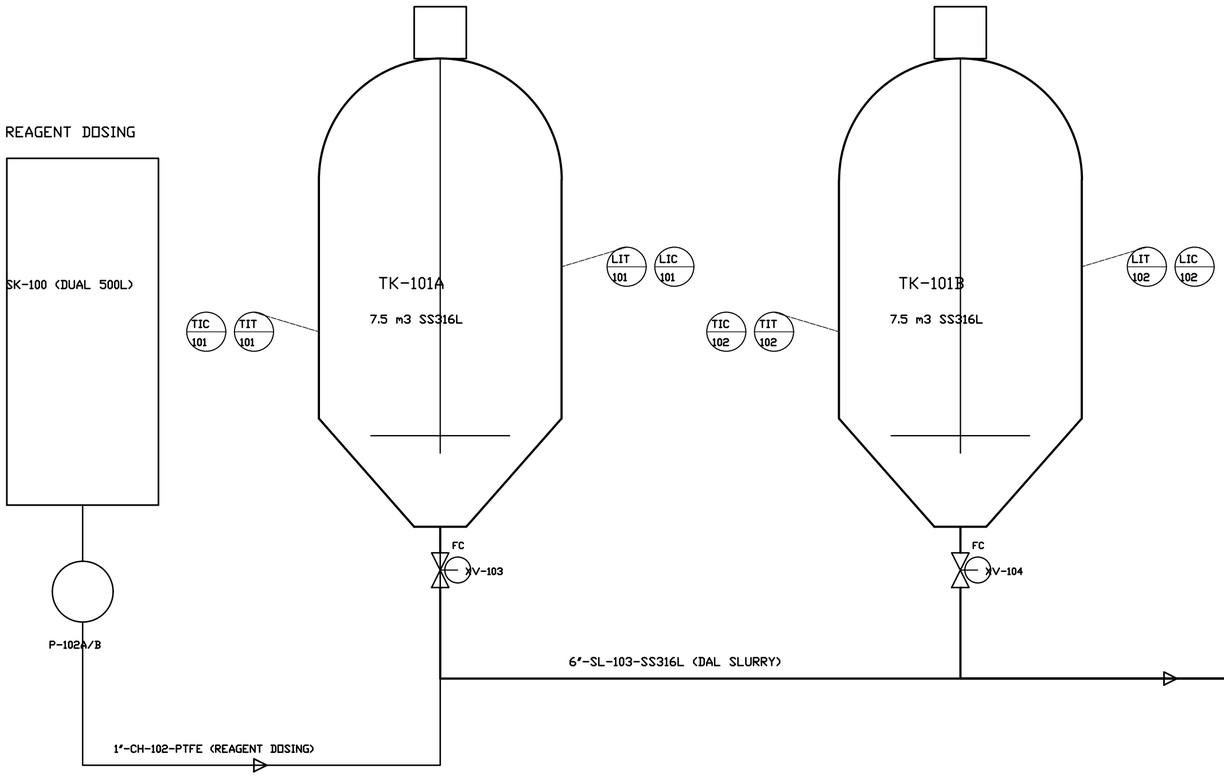
1. LOW PRESSURE STEAM DUTY (3.0 bar g saturated, 143.6 deg C, h_fg = 2,133 kJ/kg):
   - Air Heater HE-101 (Fluid Bed Dryer): 512.4 kWth -> Steam Consumption: 864.5 kg/h.
   - Triple-Effect Calandria EV-101A: 285.6 kWth -> Motive Steam Consumption: 481.8 kg/h.
   - Total Continuous Steam Demand: 1,346.3 kg/h (1.35 TPH @ design load).
   - Recommended Boiler Capacity: 2.0 TPH packaged biomass / dual-fuel boiler.
2. ELECTRICAL POWER DEMAND (415V, 3-Phase, 50 Hz):
   - Total Plant Connected Load: 184.5 kW | Operating Diversity Factor: 0.70.
   - Maximum Continuous Running Demand: 129.2 kW (131.8 kVA @ 0.98 corrected PF).
   - Substation Transformer TR-01: 500 kVA (26.4% utilization, ample Phase 2 margin).
   - Standby Emergency Generator DG-01: 250 kVA (100% full plant backup).
3. COOLING WATER & COMPRESSED AIR UTILITIES:
   - Surface Condenser HE-103 Cooling Water: 45.0 m3/h @ 32 deg C supply / 38 deg C return.
   - Plant Compressed Air: 120 Nm3/h @ 7.0 bar g (Oil-free screw compressor for instrumentation).

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FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

PROCESS STREAM TABLE & MASS-ENERGY BALANCE
DISCIPLINE: PROCESS | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-002	REV: V8.0	SHEET: 2 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD



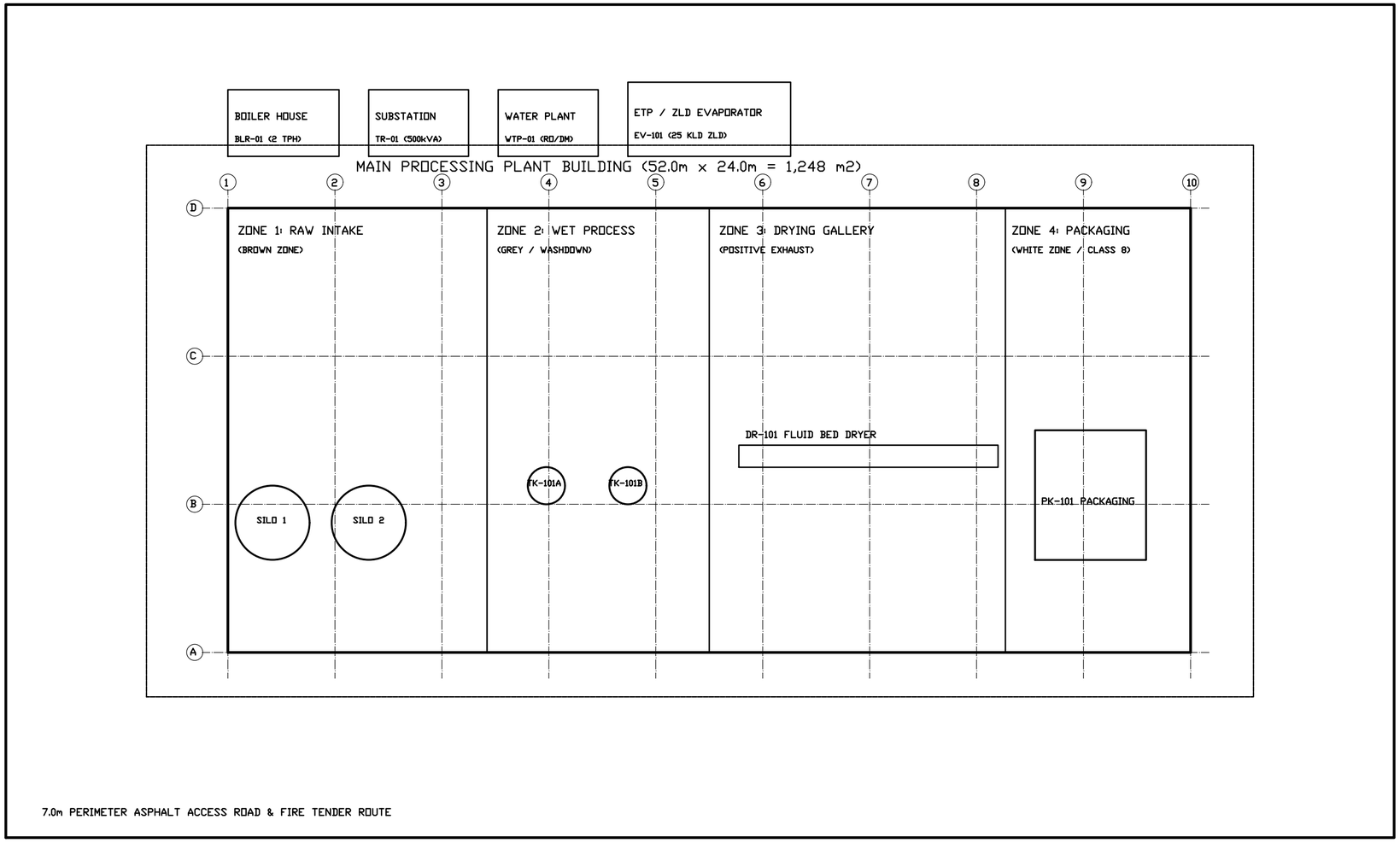
TAG NO	SERVICE	LINE / VESSEL	TRANSMITTER	CONTROLLER	FINAL CONTROL ELEMENT	INTERLOCK LOGIC & CONTROL NARRATIVE
LIC-101A	Soak Tank A Level	TK-101A	Diff Pressure LT-101A	PLC PID Loop	Pneumatic XV-101A	Auto cut-off water inlet at 85% batch level; LAH interlock alarm.
TIC-101A	Soak Tank A Temp	TK-101A	Duplex RTD TT-101A	PLC PID Loop	Modulating TCV-101A	Maintains 30.0 deg C +/- 1 deg C via steam injection coil; TAH trip.
AIC-101	Chemical Dosing pH	Dosing Line	Online pH AT-101	PLC Ratio Controller	VFD Metering Pump P-102	Paces reagent stroke volume proportional to batch water filling rate.
ZIC-103	Bottom Dump Knife	TK-101A Drain	Limit Switches ZT-103	PLC Sequence State	Air Piston Actuator XV-103	Interlocked with Dewatering Screen SC-101 run signal; prevents flood.
PIC-101	Vessel Dome Relief	TK-101A Vent	Press Trans PIT-101	Direct Acting PSV	Spring Loaded PSV-101	Protects vessel against overpressure / vacuum collapse during dump.

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15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

P&ID - RAPID HYDRATION SOAK BATTERY & DOSING SKID
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-003	REV: V8.0	SHEET: 3 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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SITE AREA ALLOCATION & MASTER METRICS:

1. Total Greenfield Plot: 3,375.0 m2 (75.0m x 45.0m).
2. Main Processing Building: 1,248.0 m2 (52.0m x 24.0m).
3. Building Ground Coverage: 36.98% (Permissible: 50.0%).
4. Floor Area Ratio (FAR): 0.42 (Permissible: 1.50).
5. External Utility Yard: 380.0 m2 (Boiler, Substation, ETP).
6. Asphalt Roads & Parking: 1,215.0 m2 (Heavy vehicle turning).
7. Green Belt & Landscaping: 532.0 m2 (15.76% plot area).
8. Finished Floor Elevation: +600 mm above finished site grade.
9. Clear Eaves Height: 8.50 m (Apex Height: 10.20 m PEB structure).
10. Structural Bays: 9 bays @ 5.78m longitudinal, 3 @ 8.0m transverse.

HYGIENIC ZONING & HVAC PRESSURIZATION SCHEDULE:

ZONE 1 (BROWN ZONE) - RAW INTAKE & STORAGE:

* Area: 336.0 m2 | Pressurization: Neutral (0 Pa) | 6 ACH Exhaust.
* Surface: Anti-abrasion floor, dust extraction hoods on chutes.

ZONE 2 (GREY ZONE) - RAPID HYDRATION & CHEMICAL TREATMENT:

* Area: 288.0 m2 | Pressurization: Neutral to -5 Pa | 10 ACH.
* Surface: Polyurethane screed floor (R11 slip), stainless drains.
* Washdown: 100% wet cleaning compatible, coved wall joints.

ZONE 3 (GREY / DRY ZONE) - THERMAL FLUID BED DRYING GALLERY:

* Area: 384.0 m2 | Pressurization: +5 Pa (Prevent raw dust ingress).
* Air Handling: Filtered dry make-up air, acoustic enclosures.

ZONE 4 (WHITE ZONE) - SECONDARY PACKAGING & FINISHED GOODS:

* Area: 240.0 m2 | Cleanliness: ISO 14644-1 Class 8 Cleanroom.
* Pressurization: +15 Pa Positive Differential (Overpressure).
* Air Filtration: Central AHU with 99.99% HEPA H14 terminal units.
* Personnel Entry: Dual-door interlocked airlock with sticky mats.

LIFE SAFETY, EGRESS & EXPLOSION COMPLIANCE:

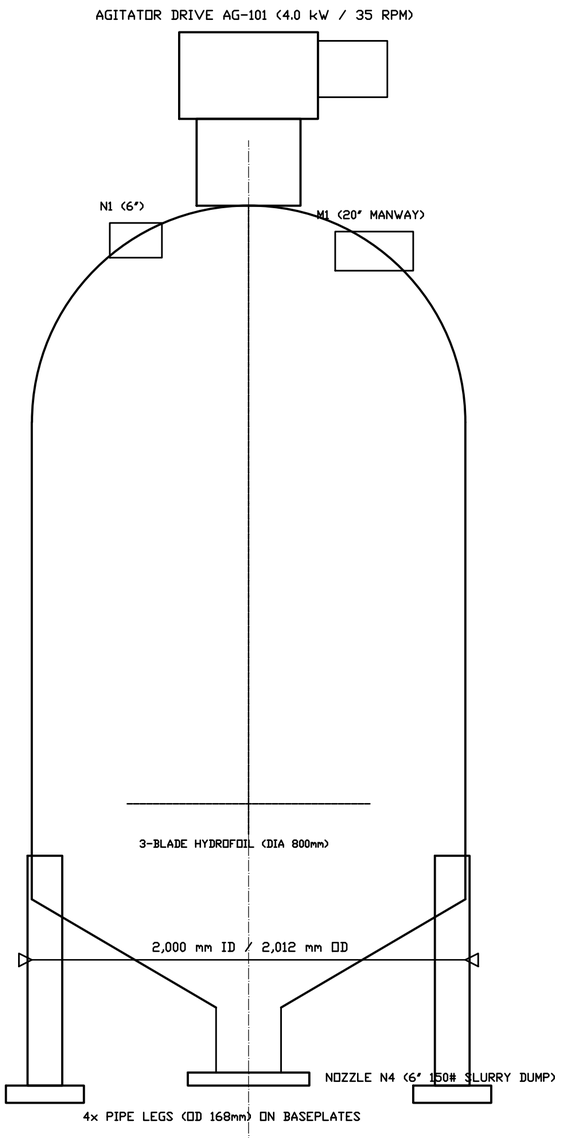
* NFPA 68 burst panel discharge ducts through roof (DR-101).
* 4x Fire egress doors with panic hardware leading to ring road.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

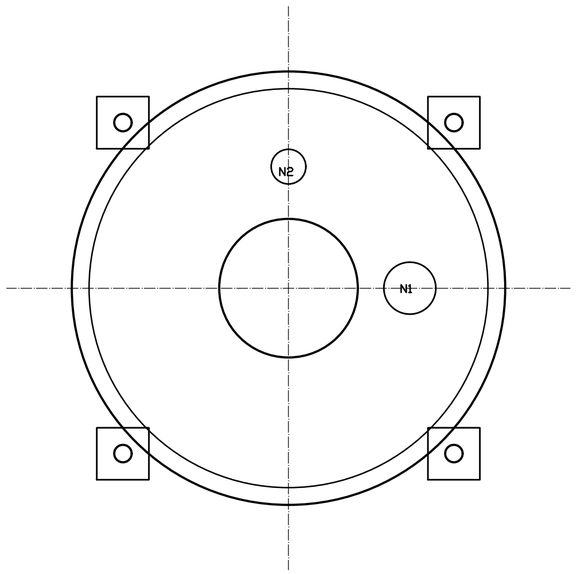
SITE MASTER PLOT PLAN & HYGIENIC ZONING
DISCIPLINE: CIVIL/LAYOUT | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-004	REV: V8.0	SHEET: 4 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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FRONT ELEVATION VIEW (SCALE 1:20)



TOP PLAN VIEW (SCALE 1:20)

MARK	SIZE	RATING	FACING	SERVICE / DESCRIPTION
N1	6" NB	150# ANSI	RF WNRF	Raw Dal Grain Infeed Chute
N2	2" NB	150# ANSI	RF WNRF	Process Soft Water Inflow
N3	1" NB	150# ANSI	RF WNRF	Chemical Reagent Dosing Inflow
N4	6" NB	150# ANSI	RF WNRF	Soaked Dal Slurry Dump to Screen
N5	2" NB	150# ANSI	RF WNRF	CIP Spray Ball Supply Header
N6	3" NB	150# ANSI	RF WNRF	Atmospheric Vent / Pressure Relief
M1	20" NB	150# ANSI	RF Davit	Inspection & Maintenance Manway

MECHANICAL DESIGN DATA & FABRICATION SPECIFICATION:

1. DESIGN CODE: ASME Boiler & Pressure Vessel Code Sec VIII Div 1 (Reference).

2. WORKING CAPACITY: 7.50 m3 | GROSS INTERNAL VOLUME: 8.45 m3.

3. DESIGN PRESSURE: Full Atmospheric (FV / 0.50 bar g Hydrostatic Liquid Head).

4. DESIGN TEMPERATURE: 65.0 deg C | OPERATING TEMPERATURE: 30.0 deg C +/- 1 deg C.

5. SHELL METALLURGY: AISI SS316L (6.0 mm nominal thk, 2B sanitary finish).

6. HEAD METALLURGY: AISI SS316L (8.0 mm thk, 2:1 Ellipsoidal seamless formed).

7. CONICAL HOPPER: AISI SS316L (6.0 mm thk, 60 deg included angle with smooth knuckle).

8. INTERNAL SURFACE FINISH: Ra <= 0.8 um (180 Grit electro-polished, weld ground flush).

9. EXTERNAL FINISH: Descaled, passivated and Scotch-Brite uniform satin finish.

10. RADIOGRAPHY & NDT: Spot Radiography per ASME UW-52 (Joint Efficiency E = 0.85).

11. HYDROSTATIC TEST: Shop Hydro-tested at 1.30 x Design = 0.65 bar g for 60 min.

12. AGITATOR: SEW-Eurodrive Helical-bevel geared motor, 4.0 kW, 35 RPM output.

13. IMPELLER: 3-Blade high-efficiency hydrofoil (Dia 800 mm, AISI SS316L solid cast).

14. CIP SYSTEM: 2x 360 deg Rotary spray balls (SS316L) covering 100% internal surface.

15. VESSEL WEIGHTS: Tare Empty Weight: 1,850 kg | Full Water-Filled Weight: 9,350 kg.

16. FOUNDATION LOADS: 4x Support legs, vertical load 23.4 kN/leg, anchor bolts 4x M20 SS304.

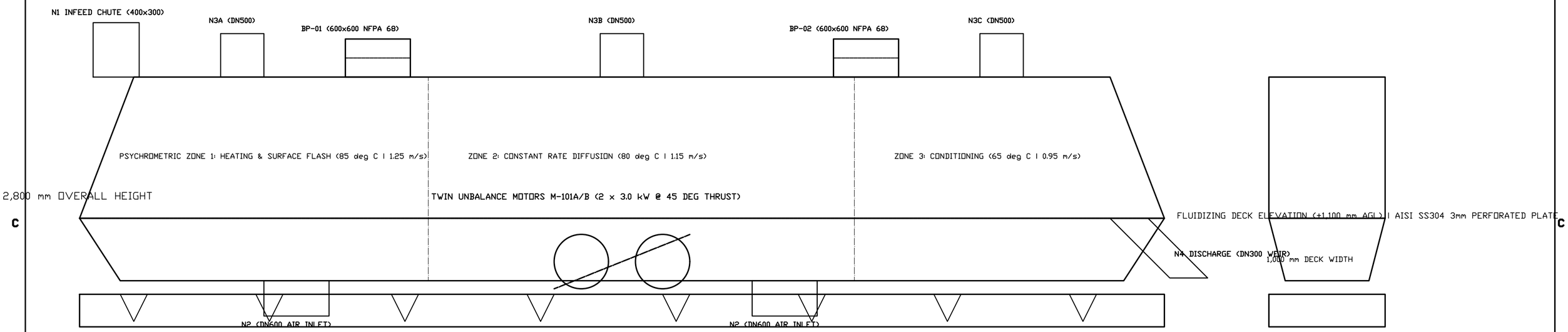
17. CERTIFICATION: Manufacturer's Data Report Form U-1 / EN 10204 3.1 Mill Test Certificates.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GENERAL ARRANGEMENT - SOAK VESSEL TK-101
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-007	REV: V8.0	SHEET: 7 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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MARK	SERVICE	SIZE	RATING	FACING	LOCATION
N1	Moist Dal Infeed	400x300	FLANGED	FLAT FACE	Top Hood Infeed End
N2A/B	Hot Air Supply	2x DN600	150#	RAISED FACE	Bottom Plenum Underside
N3A-C	Exhaust Air Outlets	3x DN500	150#	FLAT FACE	Top Hood Crown
N4	Dry Dal Discharge	DN300	150#	RAISED FACE	Discharge End Retention Weir
N5	Plenum Fines Drain	DN100	150#	RAISED FACE	Plenum Invert Low Point
M1-M4	Inspection Doors	4x 500x400	HINGED	FOOD GASKET	Drive & Non-Drive Sides
BP-01/02	NFPA 68 Burst Panels	2x 600x600	SS COMPOSITE	ROOF DUCTED	Expansion Hood Crown

MECHANICAL, VIBRATION DYNAMICS & NFPA 68 SAFETY SPECIFICATION:

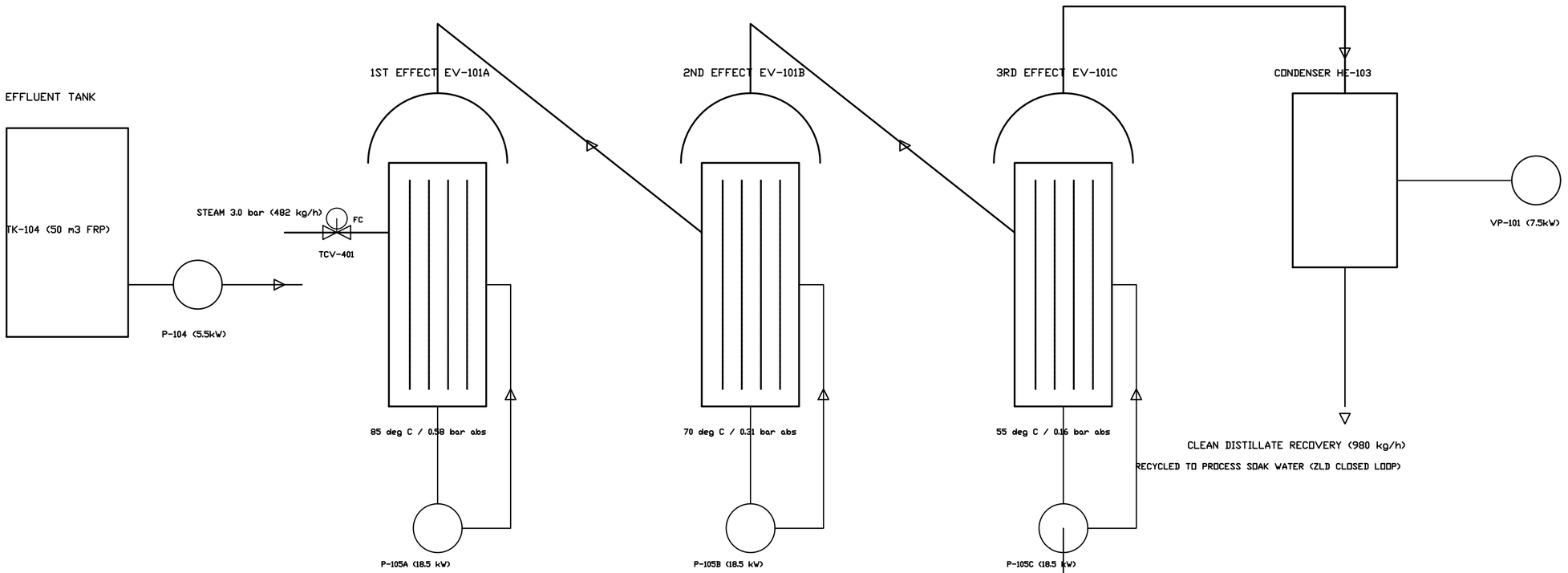
1. Equipment Tag: DR-101 | Service: Continuous Vibrating Fluid Bed Dal Conditioning & Drying.
2. Effective Deck Dimensions: 14,000 mm Length x 1,000 mm Width (Effective Area: 14.0 m2).
3. Material of Construction: All product contact parts SS304 (Ra < 0.8 um sanitary polish). Non-contact IS2062 CS.
4. Fluidizing Air Rate: 14,150 kg/h (11,800 m3/h @ 85 deg C) | Superficial Velocity: 1.15 m/s (Umf = 0.85 m/s).
5. Water Evaporation Duty: 412.34 kg/h H2O | Thermal Heating Duty: 512.4 kWth via Steam Coil HE-101.
6. Vibration Dynamics: Stroke amplitude 3.5 mm @ 16 Hz (960 RPM); Twin unbalance motors 2 x 3.0 kW @ 45 deg.
7. Isolation Efficiency: 8x Heavy-duty helical coil springs with rubber snubbers (>96% dynamic isolation).
8. Residence Time: 18.0 - 22.0 minutes plug-flow with adjustable discharge overflow retention weir.
9. NFPA 68 EXPLOSION RELIEF: Dual 600x600 mm composite rupture panels (Pred = 0.2 bar g, Pstat = 0.1 bar g).
   - Sized for St-1 Agricultural Pulse Dust (Kst = 85 bar-m/s, Pmax = 7.8 bar g) with ducted vent to building exterior.
10. Safety Hold: Final dynamic balancing, spring resonance validation and FAT testing required prior to site release.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GENERAL ARRANGEMENT - FLUID BED DRYER DR-101
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-008	REV: V8.0	SHEET: 8 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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ZLD TRIPLE-EFFECT EVAPORATOR OPERATING SPECIFICATIONS & INSTRUMENTATION:

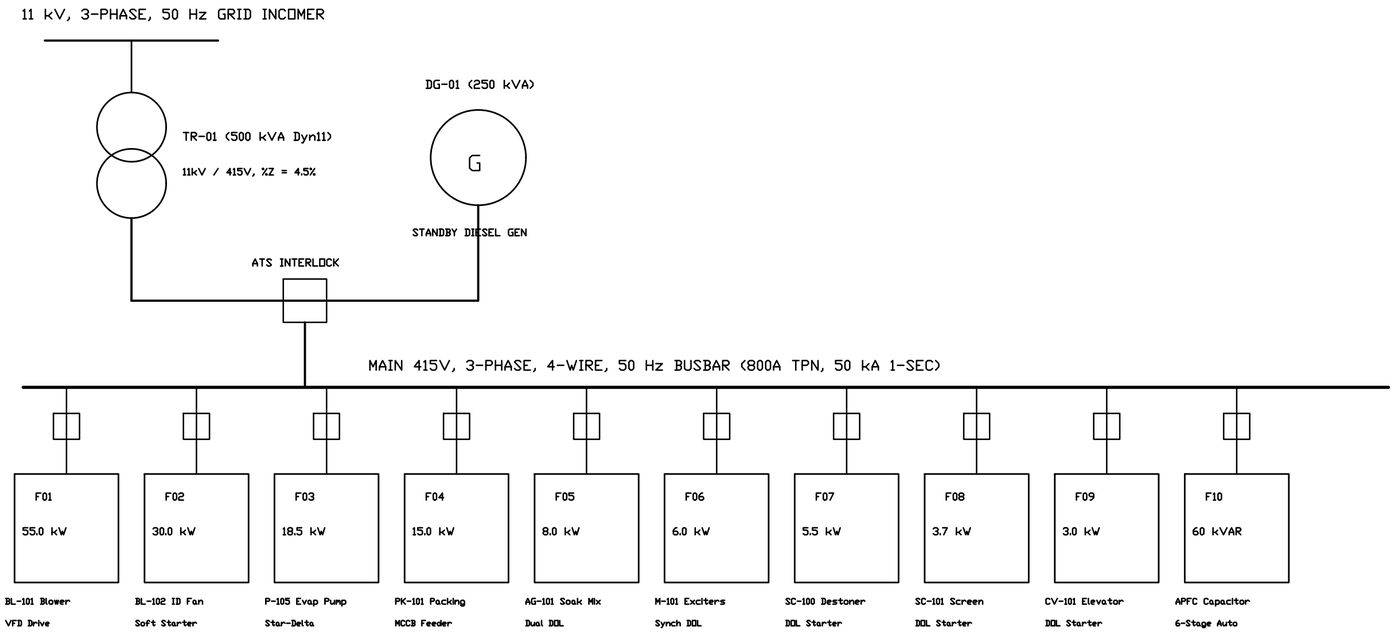
1. Feed Rate: 1,018.75 kg/h spent soak liquor @ 0.63 wt% dissolved organic/mineral solids.
2. Evaporation Capacity: 980.0 kg/h clean distillate recovery (96.2% volumetric water recovery rate).
3. Steam Economy: 2.38 kg H2O evaporated per kg 3.0 bar g motive steam supplied (0.42 kg steam / kg H2O).
4. Distillate Quality: Total Dissolved Solids (TDS) < 25 ppm, Conductivity < 40 uS/cm (Recycled directly to soak process).
5. Instrumentation: Full ISA-5.1 control loops (TIC-401 steam feed, LIC-401/402 inter-effect, DIC-403 density bleed).
6. Concentrate Discharge: 38.75 kg/h heavy slurry (40% total solids) to solar crystallization pan.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

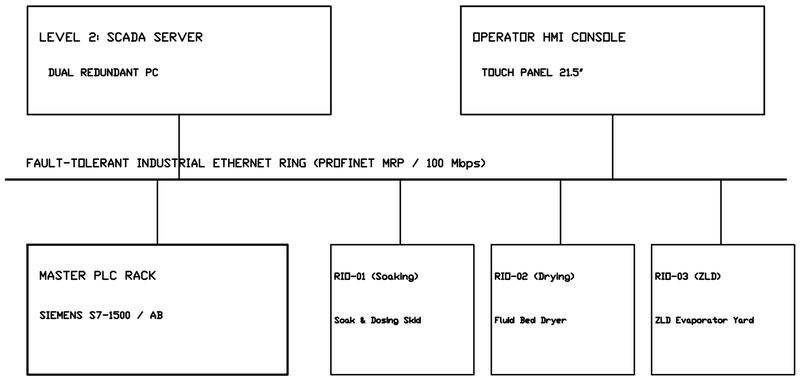
P&ID - TRIPLE-EFFECT ZLD WASTEWATER EVAPORATOR
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-010	REV: V8.0	SHEET: 10 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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INDUSTRIAL SCADA & PLC AUTOMATION ARCHITECTURE (ISA-95 / IEC 62443):



LEVEL 0 FIELD INTERFACE: HART 4-20mA Analog Loops, Modbus RS-485 Power Meters, VFDs on PROFINET

SAFETY INTEGRITY: Dual E-Stop Safety Relays (Cat 4 / SIL 3) hardwired to MCC Main Incomer Trips

ELECTRICAL LOAD SUMMARY, MOTOR FEEDER SIZING & SWITCHGEAR SCHEDULE:

1. TOTAL CONNECTED LOAD: 184.5 kW | DIVERSITY FACTOR: 0.70 | PEAK RUNNING DEMAND: 129.2 kW (131.8 kVA @ 0.98 PF).

2. POWER FACTOR CORRECTION: 60 kVAR Automatic APFC Bank maintains bus power factor at >= 0.98 lag (Penalty free).

3. TRANSFORMER SIZING: 500 kVA, 11kV/415V, Dyn11, ONAN cooling. Loading: 26.4% (Substantial headroom for Phase 2).

4. STANDBY GENERATION: 250 kVA (200 kW @ 0.8 PF) Diesel Generator with automatic mains failure (AMF) and ATS interlock.

5. UNINTERRUPTIBLE POWER SUPPLY (UPS): 10 kVA Online industrial true-sine UPS powering PLC, SCADA, and safety instruments.

6. SHORT CIRCUIT RATING: 50 kA for 1 second fault withstand rating on all MCC switchgear, busbars, and circuit breakers.

7. EARTHING & PROTECTION: Combined TN-S system with dedicated chemical copper earthing pits (<1.0 ohm resistance).

8. LIGHTNING PROTECTION: Early Streamer Emission (ESE) air terminal per IS/IEC 62305 with 45m protection radius.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

ELECTRICAL SINGLE-LINE DIAGRAM & AUTOMATION SCADA TOPOLOGY
DISCIPLINE: ELECTRICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-011	REV: V8.0	SHEET: 11 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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